

Project 1: TestFire Login Security Analysis

Objective:

To understand how a web application's login process works by analysing HTTP requests, responses, cookies, sessions, and authentication using Browser Developer Tools.

Target Application:

TestFire (Altoro Mutual) – <https://testfire.net>

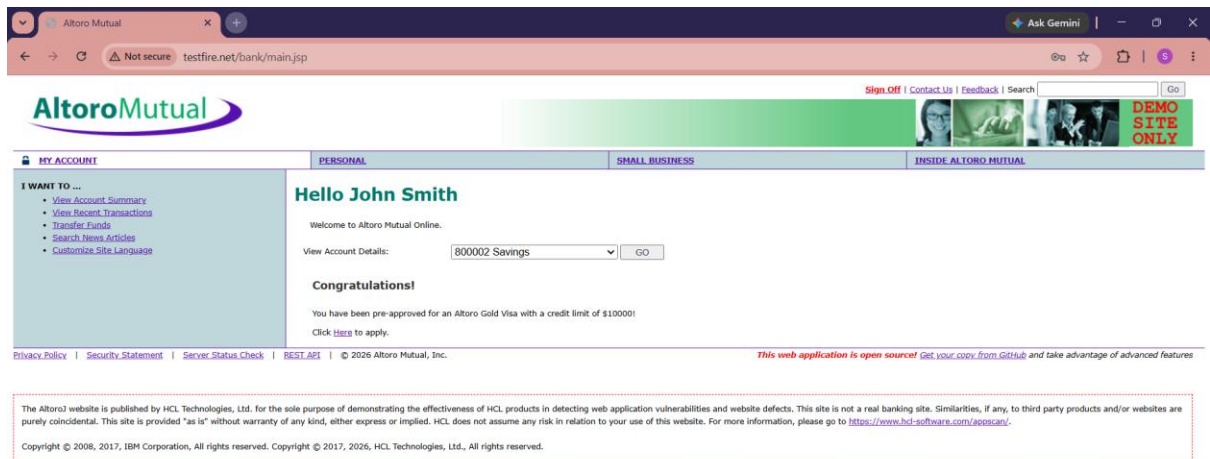
Tools Required

- Google Chrome / Mozilla Firefox
- Browser Developer Tools (F12)
- Internet Connection

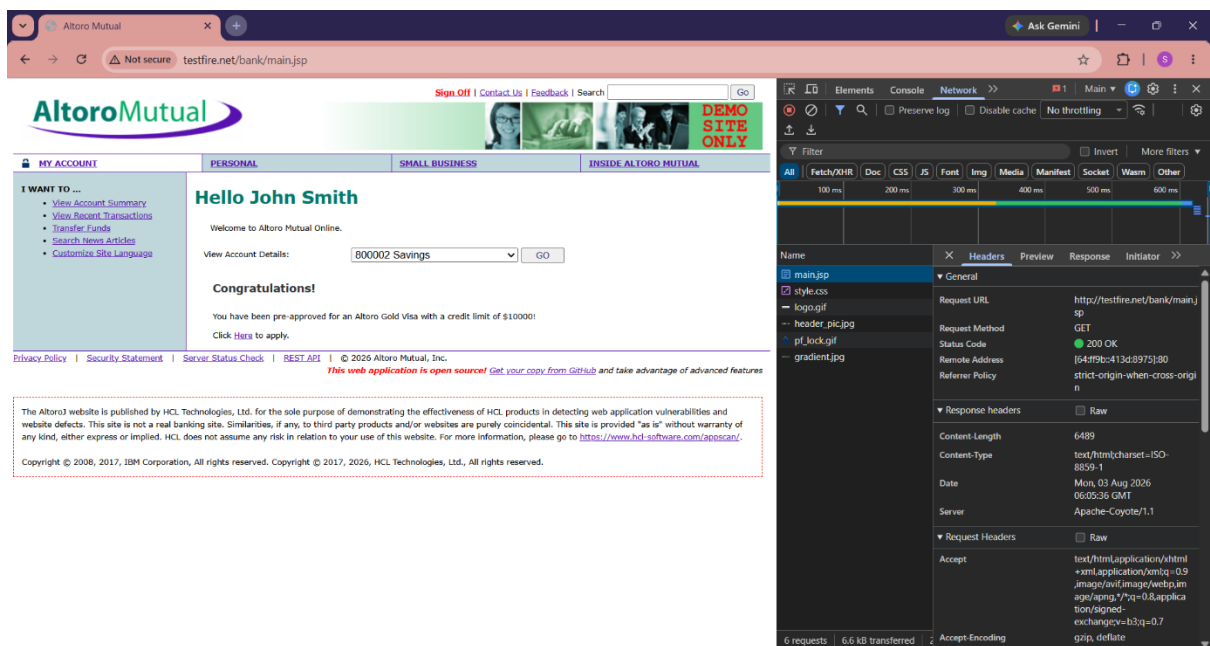
Procedure

1. Open the TestFire (Altoro Mutual) website.
2. Navigate to the **Sign In** page.
3. Press **F12** to open Browser Developer Tools.
4. Select the **Network** tab.
5. Log in using the provided demo credentials.
6. Observe the login request and record the Login URL, HTTP Method, Request Headers, Response Headers, Request Parameters, and HTTP Status Code.
7. Check the **Application/Storage** → **Cookies** section and identify the session cookie (**JSESSIONID**).
8. Log out of the application and verify that the session is terminated.

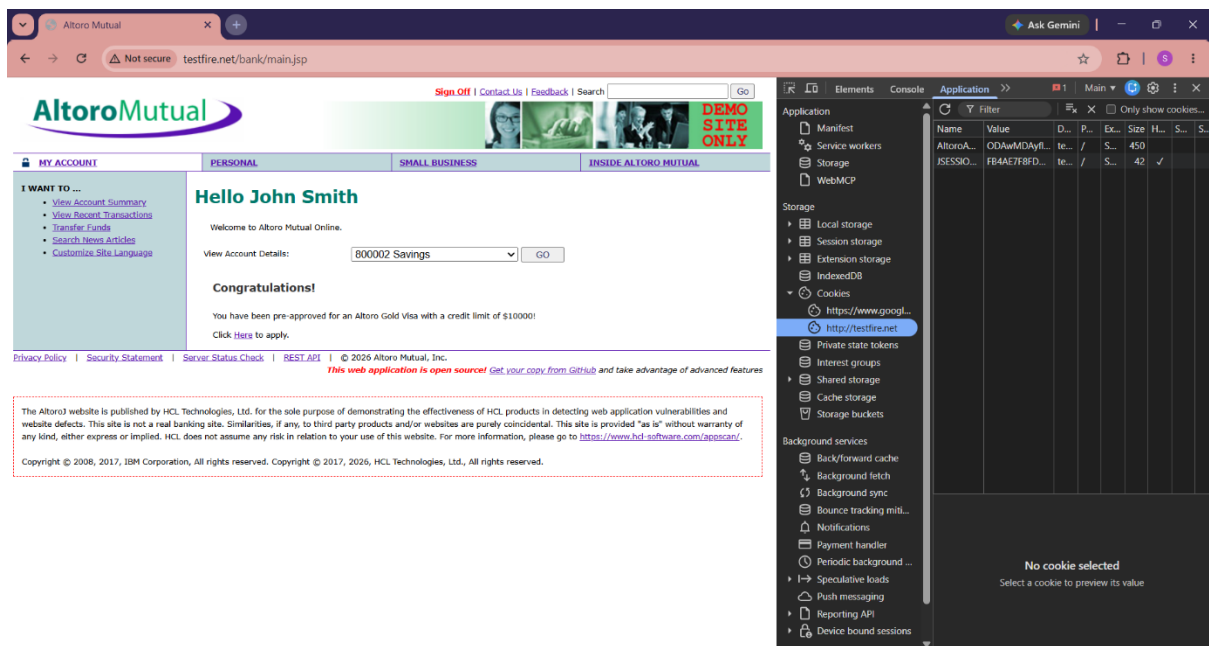
Login Successful (Hello John Smith)



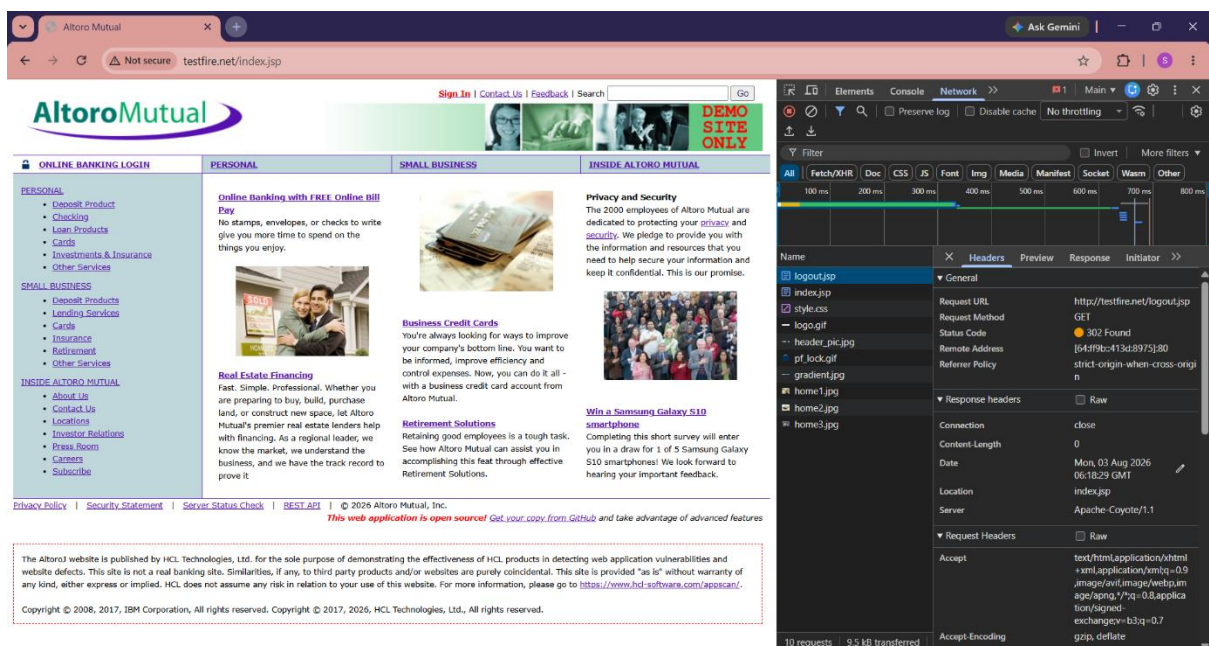
Login Request – Network Headers (main.jsp)



Cookies (JSESSIONID)



Logout Request – Network Headers (logout.jsp)



Authentication

Authentication is the process of verifying the identity of a user by checking valid login credentials such as username and password.

Authorization

Authorization is the process of determining what resources or actions an authenticated user is permitted to access after successful login.

Session

A session is a temporary connection between the client and the server that stores user information after successful authentication.

Cookie

A cookie is a small piece of data stored in the browser to maintain the user's session and preferences. In this project, the session cookie identified was **JSESSIONID**.

Security Observations

1. The application creates a **JSESSIONID** session cookie after successful login.
2. Browser Developer Tools can be used to inspect HTTP requests and responses.
3. The login request uses the HTTP protocol to communicate with the server.
4. Logging out redirects the user through **logout.jsp**, which terminates the active session.

Result

Successfully analysed the login process of the TestFire web application, observed HTTP requests and responses, identified the session cookie (**JSESSIONID**), and verified that the session was terminated after logout.